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Quantum Computing Applications in Financial Modeling and Portfolio Optimization

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Abstract— In this decade the classical computers can surpass the computational power anticipated using quantum computers, bringing transformative advancements across multiple industries, with finance expected to be the first to reap the benefits. Notably, the quantum computer poised in the financial sector leveraging medium and long terms usually but sometimes in short term. The present study involves the quantum computing applications in financial modelling and portfolio optimization using classical convex optimization algorithmthat focus on the current advancement of quantum computationsin financial modelling. The key areas focused by quantum computing includes machine learning optimization and stochastic modelling. It explodes the quantum computing techniques adopted with solutions that can enhance the problem-solving skills in finance sector like fraud detection, natural language processing, portfolio optimization, risk modelling, derivative pricing that improves both efficiency and accuracy. Additionally, we examine the practicality of implementing these algorithms on near-term quantum computers across different hardware architectures and highlight their relevance to a diverse range of financial use cases.

Keywords—Quantum Computing Applications, Financial Modeling, Portfolio Optimization

I. Introduction

The fundamental principle of storing and processing of information in different method when compared to classical computers is referred as quantum computation. Difference arises because the rules of quantum mechanics is followed by the quantum information when it is not followed by the classical mechanics.Typically, the effect of quantum mechanics emerges only through extreme small scales whereas it should be isolated carefully from the quantum systems from their surroundings to prevent interference. However, maintaining these conditions presents significant challenges in building a functional quantum computer[1].

In recent years, quantum computing has attracted significant interest, not only within physics and computer science but also across various application domains. One of the most promising areas for its impact is finance, where complex and computationally demanding problems are abundant. Many of these challenges are well-suited to leverage the advantages offered by quantum computing, making finance a key sector poised to benefit from its capabilities[2]. In the financial sector, quantum computing is being explored across various areas, making it challenging to obtain a clear and comprehensive understanding of its full potential. The intersection of quantum computing and finance involves a wide range of applications, from portfolio optimization to risk assessment, each leveraging quantum algorithms in different ways. As a result, gaining a coherent overview of how quantum computing can transform financial processes requires navigating through diverse research and implementation efforts[3]. The authors explore key topics such as Monte Carlo simulation, optimization, and machine learning. However, as all contributing authors are IBM employees, their focus is primarily on approaches that utilize IBM's quantum hardware and software.

The branch of physics that analysis the behaviour of energy and matter at sub atomic and atomic scales is known as Quantum mechanics. Is different from classical mechanics, which explains the motion of large-scale objects, quantum mechanics provides a fundamental framework for describing the physical universe. Developed in the early 20th century, it was initially formulated to explain the behavior of atoms and their constituent particles. As a probabilistic theory, it does not predict exact positions or velocities of particles but rather the probability of finding them in specific states. Two of its most significant principles are superposition and entanglement, which play a crucial role in quantum phenomena[4].

The fundamental challenge in selection of assets including loans commodities bonds and stocks in optimal manner to achieve specific financial goals is referred as portfolio management. The classical approach to portfolio management was first introduced by Harry Markowitz, who developed Modern Portfolio Theory (MPT), emphasizing the balance between risk and return through diversification[5]. Markowitz's approach centered on efficient mean-variance combinations, aiming to optimize the trade-off between expected returns and risk. His work laid the foundation for a wide range of optimization problems in portfolio management, leading to various adaptations, including optimization models of single and multi-objective[6,7], single and multiple period[7,8], without and with[9,10] stochastic or deterministic variables and transactions[11,12],

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This framework, known as portfolio optimization, explores all possible asset combinations to identify the most efficient allocation that balances risk and return. It has led to diverse optimization models, including single-objective and multi-objective approaches, to address different financial goals and constraints.

Many financial applicationsrequire the capability for evaluating the broad range of outcomes with potentiality. To achieve the outcomes, the financial sectorsrely on statistical algorithms and models, the demand for highly accurate predictive computations is growing. To meet this need, several financial institutions are exploring quantum computing, which has the potential to process vast datasets and deliver faster, more precise results than classical computers. As financial institutions prepare to harness the power of quantum computing, they anticipate significant advantages. This section explores financial use cases that are expected to benefit from a quantum computing approach.

II. Review of Literature

A. Quantum Computing in Finance

Quantum Computing is met with a blend of wonder, skepticism and enthusiasm. The quantum computers have demonstrated the ability to outperform classical computers in specific tasks and wide range of applications remains underexplored for general-purpose problems. This study focuses On quantum computing applications to address the integrated application of quantum optimization in finance for portfolio optimization problem. We provide a detailed empirical analysis of how adjustable parameters of both a cutting-edge adiabatic quantum computer and the POP itself impact the quality of the solutions produced. Our findings indicate that certain parameters, such as chain strength and the number of reads, significantly affect the results, while others, like anneal schedule and embedding, show less influence[13].

Quantum techniques are integrated with machine learning in recent days hold transformative potential for financial portfolio optimization, enabling the maximization of returns while effectively managing risks. This study aims to push the boundaries of quantum machine learning algorithms for optimal portfolio management, introducing a new Python-based approach that outperforms current methods. The algorithm's ability to deliver impressive returns over time makes it a game-changing tool for portfolio optimization in the ever-evolving financial markets. In pursuit of further improving quantum machine learning algorithms, this research centers on the development and refinement of the QSVM algorithm. Using Python for implementation, the study addresses key factors like quantum circuit optimization, noise reduction, and the seamless integration of classical and quantum components to achieve enhanced performance[14].

The paper examines the core principles of quantum computing, including qubits, highlights, entanglement and superposition its advantages over classical computing in the context of risk assessment and mitigation. It provides an overview of traditional financial risk management approaches and explores how quantum algorithms—like quantum Monte Carlo methods and quantum annealing—can improve these strategies. The paper also identifies the challenges and obstacles to integrating quantum computing within the financial sector and outlines potential future research areas. Ultimately, it emphasizing the potential of transformativequantum computing in advancing risk management within the complex landscape of modern financial markets[15].

III. Methodology

The strategic management for the group of assets can be performed with portfolio optimization aimed at minimising the risk and maximising the returns. The methodology discusses about the essential components for solving the problem such as the characteristics of asset including correlations risks and expected returns. Next the optimization algorithm framework is introduced and the key objectives was outlined including the constraints. The financial case study as real-world example acts as the main concept for the study analysis. This case study focuses on optimizing a portfolio of outstanding loans, considering factors like Carbon footprint reduction, concentration risks and return on capital goals.

$$\min_{x \in (0,1)^n} x^T Q x$$

A. Basic Problem

The portfolio optimization algorithm includes the fundamental problem involving the assets available for the investment a set of N, represented as $P1$, $P2$, ..., PN. Each asset is characterized by its expected return μi and associated risk σi, where σi is the standard deviation of the asset's returns. The returns of these assets are interconnected, with their relationships determined by the correlation ρij, which represents the correlation between assets i and j. The vector of expected returns is represented as $\mu = (\mu i)$, and the risk matrix as $\Sigma = (\sigma ij)$, where;

$$\sigma_{ij} = \begin{cases} \sigma_i^2; if i = j \\ \rho_{ij}\sigma_i\sigma_j; if i \neq j \end{cases}$$

The binary variables like x_i is introduced in the study when chosen 1 is the asset otherwise it is zero.It gives to the optimization problem as given below;

$$min \sum_{i=1}^{N}\sum_{j=1}^{N} x_i x_j \sigma_{ij}$$

$$Such\ that \sum_{i=1}^{N} x_i = N$$

$$And\ \mu^1 x \geq R^*$$

B. Financial Case

The research study examines the banking issue as real-world example with primary objective of determining the limited resource allocation in the bank as optimal allocation The focus is to distribute the portfolio of credit loan across various countries, subsectors and industrial sectors by 2030. It is to align with the sustainability goals, vision and mission of the bank comprised in the Paris agreement outline such as the pathway involving development of climate resilience and the low greenhouse gas emission. The problem can be achieved and framed with the allocation challenge with optimal asset. Particularly, the goal is to develop the strategy

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for business growth that balances the CO2 emission, rewards and risk effectively.

Consequently, the studyfocus on a portfolio to understand the loans as it stood in 2021, with the goal of setting a target for the composition and distribution of this portfolio by the year 2030.

$$HHI(x) = \sum_{i=1}^{N} \left(\frac{x_i}{\sum_{j=1}^{N} x_j} \right)^2$$

$$ROC(x) = \left(\frac{\sum_{i=1}^{N} x_i r_i / y_i}{\sum_{i=1}^{N} x_i c_i / y_i} \right)$$

Finally, the intensity of emission is recorded in the portfolio by 2021.

$$E = \frac{\sum_{i=1}^{N} e_i y_i}{\sum_{i=1}^{N} y_i}$$

$$\sum_{i=1}^{N} 0.76 x_i e_i \leq 0.7E \sum_{i=1}^{N} x_i$$

The total problem can be defined as;

$$min \phi HHI(x) - (1-\phi) ROC(x)$$

Under the below Constraints;

$$LB_i \leq x_i \leq UB_i \quad \forall i \epsilon \{1,2, \dots N\}$$

$$\sum_{i=1}^{N} x_i \, 0.76 x_i e_i \leq 0.7E \sum_{i=1}^{N} x_i$$

It leads to QUBO formulation;

$$\text{QUBO} = \min_{z} \lambda_1 \sum_{i=1}^{N} x_i^2 - \lambda_2 \sum_{i=1}^{N} \theta_i x_i + \lambda_3 (0.76 \sum_{i=1}^{N} x_i e_i - 0.7E \sum_{i=1}^{N} x_i$$

C. Convex optimization

The convex optimization algorithm is of conventional type which is developed long before to tackle the complex problems in portfolio optimization These range involves the convex optimization and conventional type of quadratic programming techniques to more modern heuristic algorithms. The convex optimization algorithm is given in table 1.

TABLE I. CONVEX OPTIMIZATION ALGORITHM

Step 1: $LB_{ROC} \leq ROC(x) \leq UB_{ROC}$
Step 2:LB $\leq ROC \leq$ UB
Step 3: $R \leftarrow LB_{ROC}$ to UB_{ROC} do
Step 4: $Set\ \bar{\gamma} = L$with L as large positive number
Step 5: for X$\leftarrow$ $LB\ to$ UB do
Step 6: $Create\ Column\ vextor\ with\ x_i\ as\ components$
Step 7: $Create\ \gamma\ as\ scalar\ LMI\ Variable$
Step 8: $Add\ outstanding\ loan\ constraints$
Step 9: $Add\ constraint: \gamma > 0$
Step 10: $Add\ ROC\ Constraint\ \sum_{i=1}^{N} x_i r_i / y_i$
Step 11: $Add\ emission\ constraint$
Step 12: $Add\ constraint\ HHI\ through\ linearization$
Step 13: $\begin{bmatrix} \gamma & (\times/X)^T \\ (\times/X) & I \end{bmatrix} > 0$
Step 14: Solve for $\hat{\times}$ and $\hat{\gamma}$
Step 15: If $\hat{\gamma} < \bar{\gamma}$ and set $\hat{\gamma} = \bar{\gamma}$

IV. RESULTS AND DISCUSSION

The present study sets a benchmark for solving the problem in portfolio optimization using the conventional convex optimization method as the pseudo code is explained in table I. This can be achieved with linear matrix inequality problem to solve using the SeDuMi solver and Matlab-YALMIP toolbox. It results with Pareto frontierAnd the outcomes are represented as green doubts and green diamonds as given in figure 1.

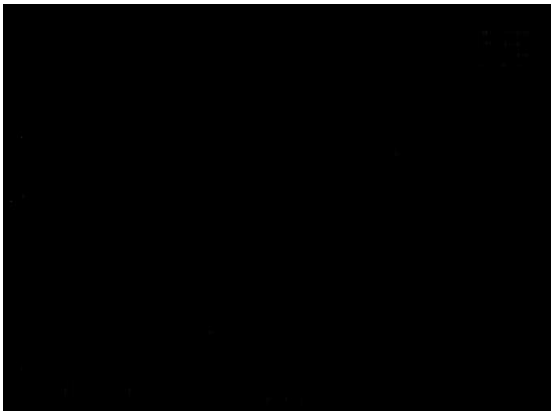

Fig. 1. Random Sampling of Portfolios

Sampling technique is used as the annealing approach to solve the QUBO problem. To demonstrate the benefits of the quantum annealing method, we compared it with naive sampling. In this simple comparison, we sampled from the entire set of possible solutions, where for each asset i in 2030, the investment amount was either the lower bound ($xi = LBi$) or the upper bound ($xi = UBi$). This approach resulted in 252,252 solutions.The Orange and blue dots represent the random portfolio solution obtained for the study and the emission constraint can be satisfied with the single orange

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dot. The Red Pareto Frontier is far from the blue dots in which the green pareto frontier is distant from the single orange dot.

Then the study applies simulated annealing for generation of solution to the optimization problem from formulation of QUBO to determine the desired outcomes as yieldedthe algorithm for classical probabilistic optimization is referred as simulated annealing which is inspired from the metallurgy process. It is designed for finding the approximate solution from using various search problems and optimization techniques. The used algorithm explores the random moves with the solution space iteratively by rejecting or accepting them depending upon the probabilistic criterion.

Fig. 2. Simulated annealing of QUBO

The findings of the experiment demonstrate the quantum computing applications in financial sector for modelling to find the solutions to satisfy the constraints related to portfolio management.Simulated annealing outperforms the quantum annealing technique based on finding solutions closer to Pareto Frontier. The study suggests the further optimization with parameters of QUBO along with solvers for deep understanding could lead to even more better results Notably the proposed approach generates broad range of solutions for the portfolio optimization problem while the limited set of Pareto Solutions is produced by the original method From this the study proves that valuable practise as it uncovered the diverse array of problem solution beyond the optimal set from original method providing alternative pathways and the wealth of insights.

V. Conclusion

To conclude the present study involves portfolio optimization using quantum computing focusing specifically with multi objective optimization algorithm. The classical challenges such as quadratic programming problems are addressed for portfolio management with variables of Binary decision and a limitation for traditional solvers The promising alternative solutions are offered by the quantum computing to overcome these obstacles utilising simulated and quantum annealing using quantum approximate optimization algorithm for effective handling of complex problems in Optimization.The key areas focused by quantum computing includes machine learning optimization and stochastic modelling. It explodes the quantum computing techniques adopted with solutions that can enhance the problem-solving skills in finance sector like fraud detection, natural language processing, portfolio optimization, risk modelling, derivative pricing that improves both efficiency and accuracy. Additionally, we examine the practicality of implementing these algorithms on near-term quantum computers across different hardware architectures and highlight their relevance to a diverse range of financial use cases. The QUBO Framework is used to tackle the different types of problems to find the effective solutions near pareto frontier.

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